

confocal_lacerate

```
# Build 6 individual panels + assemble into a 2x3 figure  
# Panels: (EdU, Caspase) x (Apo, Inoc, Sym)
```

```
library(tidyverse)  
library(dplyr)  
library(stringr)  
library(rstatix)  
library(patchwork)  
library(car)  
library(emmeans)  
library(multcomp)  
library(multcompView)  
library(tibble)
```

```
rm(list = ls())  
#graphics.off()
```

```
select <- dplyr::select  
filter <- dplyr::filter  
mutate <- dplyr::mutate
```

```
getwd()
```

```
## [1] "/Users/junbc/Downloads"
```

```
setwd("~/Documents/GitHub/heating_lacerates_final")
```

```
raw_confocal_data <- read_csv("data/Lacerate_EdU_Caspase.csv")
```

```
confocal_data <- raw_confocal_data %>%  
  filter(True_Channel %in% c("EdU", "Caspase")) %>%  
  mutate(  
    marker = factor(True_Channel, levels = c("EdU", "Caspase")),  
    time = factor(Time, levels = c("2dpl", "5dpl", "8dpl", "14dpl")),  
    treatment = as.character(Treatment),  
    state = case_when(  
      str_detect(treatment, regex("Apo", ignore_case = TRUE)) ~ "Apo",  
      str_detect(treatment, regex("Inoc", ignore_case = TRUE)) ~ "Inoc",  
      str_detect(treatment, regex("Sym", ignore_case = TRUE)) ~ "Sym",  
      TRUE ~ NA_character_  
    ),  
    heat = case_when(  
      str_detect(treatment, regex("HS|Heat", ignore_case = TRUE)) ~ "Heat",  
      str_detect(treatment, regex("Control", ignore_case = TRUE)) ~ "Control",  
      TRUE ~ NA_character_  
    )  
  )
```

```

),
state = factor(state, levels = c("Apo", "Inoc", "Sym")),
heat = factor(heat, levels = c("Control", "Heat")),
percentage = Percentage
) %>%
drop_na(marker, state, heat, time, percentage)

sample_size_table_wide <- confocal_data %>%
count(marker, state, time, heat) %>%
pivot_wider(names_from = heat, values_from = n)

print(sample_size_table_wide)

```

```

## # A tibble: 22 x 5
##   marker state time Control Heat
##   <fct> <fct> <fct>   <int> <int>
## 1 EdU    Apo   2dpl      3      6
## 2 EdU    Apo   5dpl      3      5
## 3 EdU    Apo   8dpl      5      4
## 4 EdU    Apo  14dpl      5      4
## 5 EdU    Inoc  5dpl      4      5
## 6 EdU    Inoc  8dpl      4      5
## 7 EdU    Inoc 14dpl      7      6
## 8 EdU    Sym   2dpl      5      6
## 9 EdU    Sym   5dpl     10      8
## 10 EdU   Sym   8dpl      6      8
## # i 12 more rows

```

```

# n per panel and treatment for p6
p6_n <- confocal_data %>%
count(marker, state, time, heat) %>%
arrange(marker, state, time, heat)

print(p6_n, n = Inf)

```

```

## # A tibble: 44 x 5
##   marker state time heat      n
##   <fct> <fct> <fct> <fct>   <int>
## 1 EdU    Apo   2dpl Control    3
## 2 EdU    Apo   2dpl Heat       6
## 3 EdU    Apo   5dpl Control    3
## 4 EdU    Apo   5dpl Heat       5
## 5 EdU    Apo   8dpl Control    5
## 6 EdU    Apo   8dpl Heat       4
## 7 EdU    Apo  14dpl Control    5
## 8 EdU    Apo  14dpl Heat       4
## 9 EdU    Inoc  5dpl Control    4
## 10 EdU   Inoc  5dpl Heat       5
## 11 EdU   Inoc  8dpl Control    4
## 12 EdU   Inoc  8dpl Heat       5
## 13 EdU   Inoc 14dpl Control    7
## 14 EdU   Inoc 14dpl Heat       6

```

```
## 15 EdU      Sym  2dpl  Control    5
## 16 EdU      Sym  2dpl  Heat       6
## 17 EdU      Sym  5dpl  Control   10
## 18 EdU      Sym  5dpl  Heat       8
## 19 EdU      Sym  8dpl  Control    6
## 20 EdU      Sym  8dpl  Heat       8
## 21 EdU      Sym 14dpl  Control    5
## 22 EdU      Sym 14dpl  Heat       9
## 23 Caspase Apo 2dpl  Control    3
## 24 Caspase Apo 2dpl  Heat       6
## 25 Caspase Apo 5dpl  Control    4
## 26 Caspase Apo 5dpl  Heat       5
## 27 Caspase Apo 8dpl  Control    6
## 28 Caspase Apo 8dpl  Heat       4
## 29 Caspase Apo 14dpl Control    5
## 30 Caspase Apo 14dpl Heat       4
## 31 Caspase Inoc 5dpl  Control    4
## 32 Caspase Inoc 5dpl  Heat       6
## 33 Caspase Inoc 8dpl  Control    4
## 34 Caspase Inoc 8dpl  Heat       5
## 35 Caspase Inoc 14dpl Control    7
## 36 Caspase Inoc 14dpl Heat       5
## 37 Caspase Sym  2dpl  Control    6
## 38 Caspase Sym  2dpl  Heat       6
## 39 Caspase Sym  5dpl  Control   10
## 40 Caspase Sym  5dpl  Heat       8
## 41 Caspase Sym  8dpl  Control    6
## 42 Caspase Sym  8dpl  Heat       8
## 43 Caspase Sym 14dpl  Control    5
## 44 Caspase Sym 14dpl  Heat       9
```

```
ymax_edu <- confocal_data %>%
  filter(marker == "EdU") %>%
  summarise(max_val = max(percentage, na.rm = TRUE)) %>%
  pull(max_val)

ymax_cas <- confocal_data %>%
  filter(marker == "Caspase") %>%
  summarise(max_val = max(percentage, na.rm = TRUE)) %>%
  pull(max_val)

# add headroom for letters
ymax_edu <- ymax_edu + 5
ymax_cas <- ymax_cas + 5

make_panel <- function(data_sub, panel_title, ylab = NULL, show_legend = FALSE, y_limit = NULL) {

  fit <- lm(percentage ~ heat * time, data = data_sub)

  emm <- emmeans(fit, ~ heat | time)

  letters_df <- cld(emm, adjust = "tukey", Letters = letters) %>%
    as.data.frame() %>%
    mutate(.group = str_trim(.group))
}
```

```

# get max per timepoint for positioning
y_pos <- data_sub %>%
  group_by(time, heat) %>%
  summarise(y = max(percentage, na.rm = TRUE), .groups = "drop")

letters_df <- left_join(letters_df, y_pos, by = c("time", "heat")) %>%
  mutate(y = y + 6.5)

if (is.null(y_limit)) {
  y_max <- max(c(data_sub$percentage, letters_df$y), na.rm = TRUE) + 2
} else {
  y_max <- y_limit
}

ggplot(data_sub, aes(x = time, y = percentage, color = heat)) +
  geom_jitter(
    position = position_jitterdodge(jitter.width = 0.12, dodge.width = 0.75),
    size = 2.2,
    alpha = 0.9
  ) +
  geom_boxplot(
    aes(group = interaction(time, heat)),
    outlier.shape = NA,
    width = 0.6,
    fill = NA,
    linewidth = 0.8,
    position = position_dodge(width = 0.75)
  ) +
  geom_text(
    data = letters_df,
    aes(x = time, y = y, label = .group, group = heat),
    position = position_dodge(width = 0.75),
    inherit.aes = FALSE,
    fontface = "bold",
    size = 4,
    color = "black"
  ) +
  scale_color_manual(values = c("Control" = "blue", "Heat" = "red")) +
  coord_cartesian(ylim = c(0, y_max), clip = "off") +
  labs(title = panel_title, x = NULL, y = ylab, color = NULL) +
  theme_classic(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = if (show_legend) "top" else "none",
    plot.margin = margin(8, 8, 8, 8)
  )
}

# ---- EdU (shared scale)
p_edu_apo <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Apo"),
  "EdU - Apo",

```

```

    "EdU-positive area (%)",
    y_limit = ymax_edu
  )

p_edu_inoc <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Inoc"),
  "EdU - Inoc",
  y_limit = ymax_edu
)

p_edu_sym <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Sym"),
  "EdU - Sym",
  show_legend = TRUE,
  y_limit = ymax_edu
)

# ---- Caspase (shared scale)
p_cas_apo <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Apo"),
  "Caspase - Apo",
  "Caspase-positive area (%)",
  y_limit = ymax_cas
)

p_cas_inoc <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Inoc"),
  "Caspase - Inoc",
  y_limit = ymax_cas
)

p_cas_sym <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Sym"),
  "Caspase - Sym",
  show_legend = TRUE,
  y_limit = ymax_cas
)

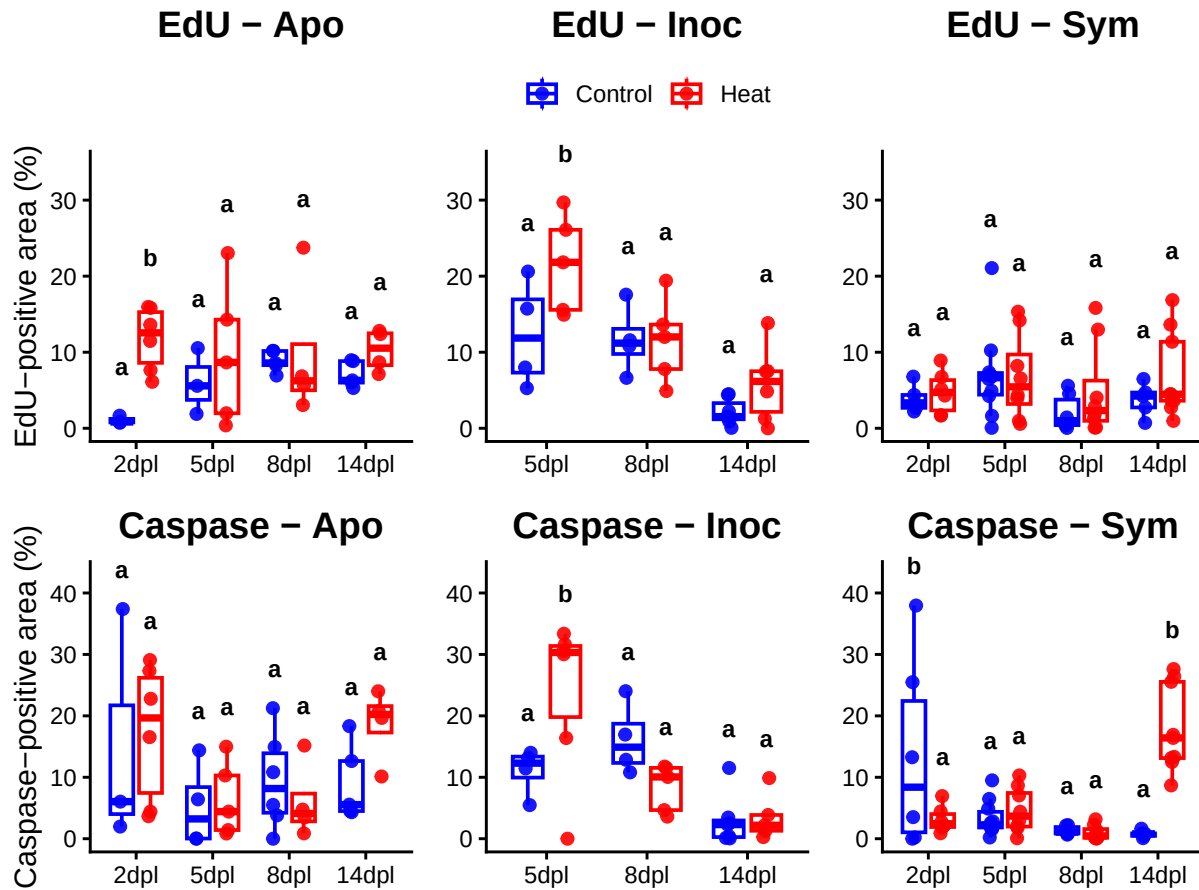
p_edu3 <- (p_edu_apo | p_edu_inoc | p_edu_sym) +
  plot_layout(guides = "collect") &
  theme(legend.position = "top")

p_cas3 <- (p_cas_apo | p_cas_inoc | p_cas_sym) +
  plot_layout(guides = "collect") &
  theme(legend.position = "top")

p6 <- (p_edu_apo | p_edu_inoc | p_edu_sym) /
  (p_cas_apo | p_cas_inoc | p_cas_sym) +
  plot_layout(guides = "collect") &
  theme(legend.position = "top")

print(p6)

```



```
ggsave(
  filename = "Fig6_confocal.png",
  plot = p6,
  path = "figs",
  device = "png",
  width = 11,
  height = 8,
  units = "in",
  dpi = 600,
  bg = "white"
)
```

```
ggsave(
  filename = "Fig6_confocal.pdf",
  plot = p6,
  path = "figs",
  device = pdf,
  width = 11,
  height = 8,
  units = "in",
  bg = "white"
)
```

```

# ---- EdU (shared scale)
p_edu_apo <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Apo"),
  "EdU - Apo",
  "EdU-positive area (%)",
  y_limit = ymax_edu
)

p_edu_inoc <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Inoc"),
  "EdU - Inoc",
  y_limit = ymax_edu
)

p_edu_sym <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Sym"),
  "EdU - Sym",
  show_legend = TRUE,
  y_limit = ymax_edu
)

# ---- Caspase (shared scale)
p_cas_apo <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Apo"),
  "Caspase - Apo",
  "Caspase-positive area (%)",
  y_limit = ymax_cas
)

p_cas_inoc <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Inoc"),
  "Caspase - Inoc",
  y_limit = ymax_cas
)

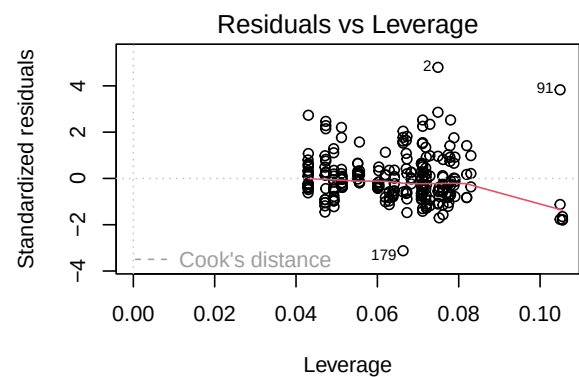
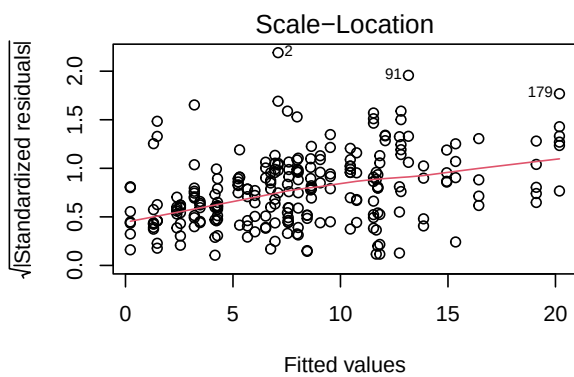
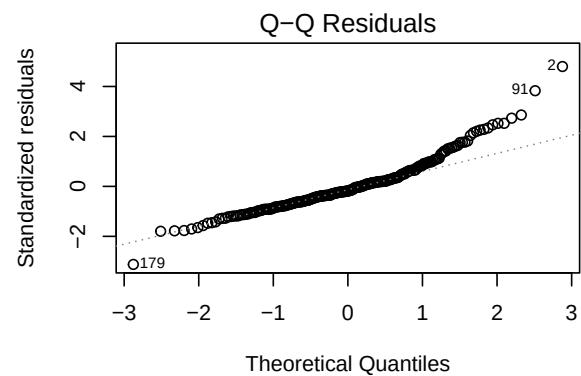
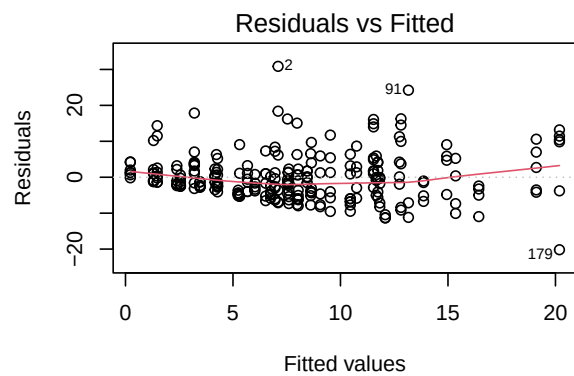
p_cas_sym <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Sym"),
  "Caspase - Sym",
  show_legend = TRUE,
  y_limit = ymax_cas
)

fit_global <- lm(
  percentage ~ marker + state + heat * time + state * time,
  data = confocal_data
)

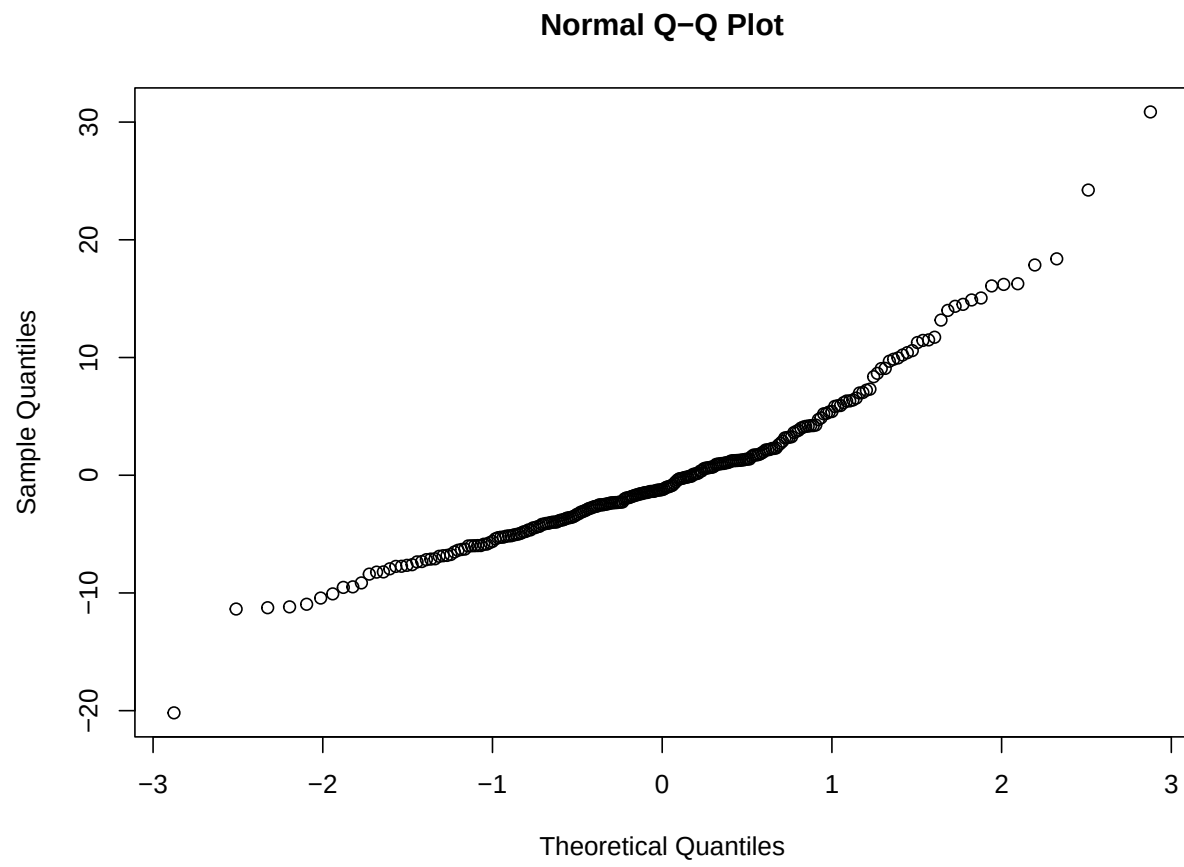
## Global model checks

par(mfrow = c(2, 2))
plot(fit_global)

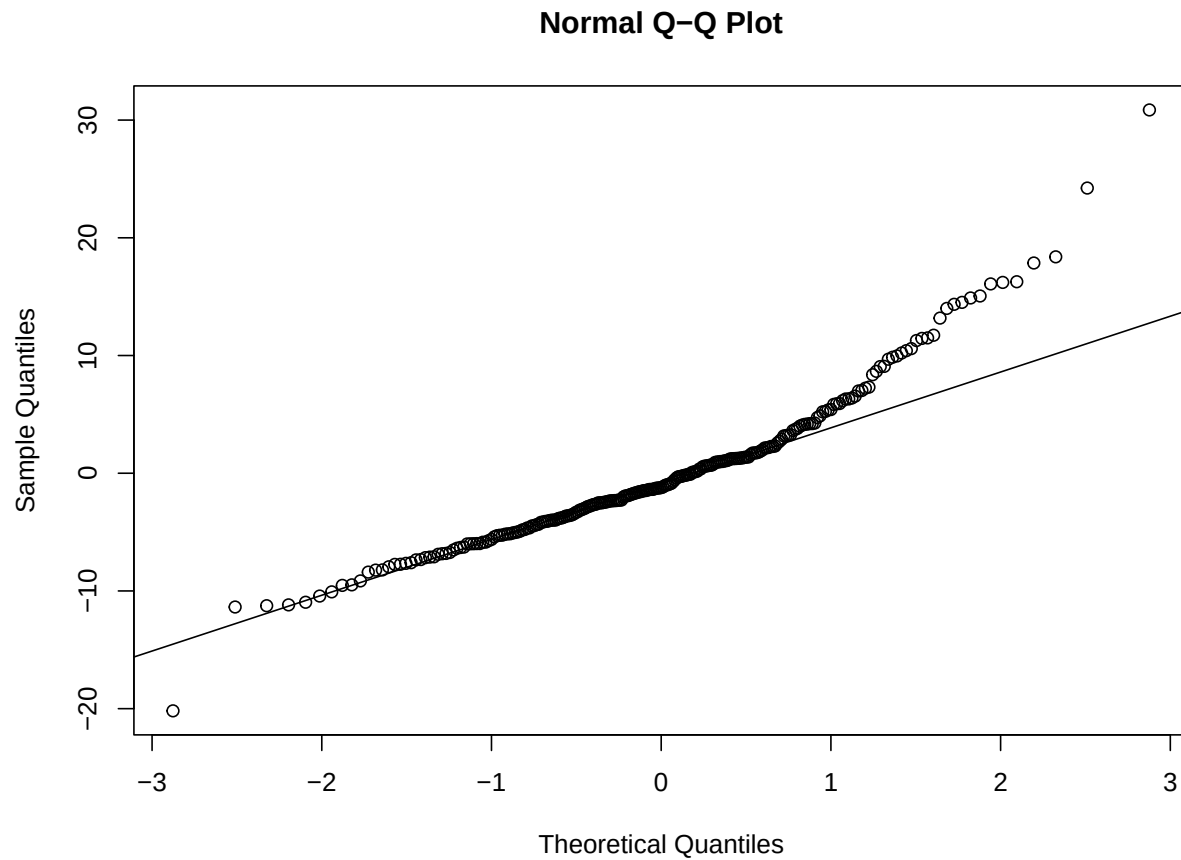
```



```
par(mfrow = c(1, 1))
qqnorm(residuals(fit_global))
```



```
qqline(residuals(fit_global))
```



```
shapiro.test(residuals(fit_global))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(fit_global)
## W = 0.93424, p-value = 4.264e-09
```

```
leveneTest(
  percentage ~ marker * state * heat * time,
  data = confocal_data
)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value    Pr(>F)
## group 43  1.9537 0.001075 **
##      205
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Type II ANOVA

```
anova_global <- car::Anova(fit_global, type = 2)

anova_global_table <- as.data.frame(anova_global) %>%
  rownames_to_column("term") %>%
  mutate(
    across(where(is.numeric), ~ signif(.x, 3))
  )
anova_global_table
```

##	term	Sum Sq	Df	F value	Pr(>F)
## 1	marker	72.3	1	1.62	2.05e-01
## 2	state	1450.0	2	16.20	2.61e-07
## 3	heat	400.0	1	8.95	3.07e-03
## 4	time	385.0	3	2.87	3.71e-02
## 5	heat:time	537.0	3	4.00	8.33e-03
## 6	state:time	2460.0	5	11.00	1.55e-09
## 7	Residuals	10400.0	233	NA	NA

```
edu_apo <- confocal_data %>%
  filter(marker == "EdU", state == "Apo") %>%
  droplevels()

fit_edu_apo <- lm(percentage ~ heat * time, data = edu_apo)
anova_edu_apo <- car::Anova(fit_edu_apo, type = 2)
emm_edu_apo <- emmeans(fit_edu_apo, ~ heat | time)
tukey_edu_apo <- pairs(emm_edu_apo, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "EdU", state = "Apo")
letters_edu_apo <- cld(emm_edu_apo, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "EdU", state = "Apo")

edu_inoc <- confocal_data %>%
  filter(marker == "EdU", state == "Inoc") %>%
  droplevels()

fit_edu_inoc <- lm(percentage ~ heat * time, data = edu_inoc)
anova_edu_inoc <- car::Anova(fit_edu_inoc, type = 2)
emm_edu_inoc <- emmeans(fit_edu_inoc, ~ heat | time)
tukey_edu_inoc <- pairs(emm_edu_inoc, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "EdU", state = "Inoc")
letters_edu_inoc <- cld(emm_edu_inoc, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "EdU", state = "Inoc")

edu_sym <- confocal_data %>%
  filter(marker == "EdU", state == "Sym") %>%
  droplevels()

fit_edu_sym <- lm(percentage ~ heat * time, data = edu_sym)
```

```

anova_edu_sym <- car::Anova(fit_edu_sym, type = 2)
emm_edu_sym <- emmeans(fit_edu_sym, ~ heat | time)
tukey_edu_sym <- pairs(emm_edu_sym, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "EdU", state = "Sym")
letters_edu_sym <- cld(emm_edu_sym, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "EdU", state = "Sym")

cas_apo <- confocal_data %>%
  filter(marker == "Caspase", state == "Apo") %>%
  droplevels()

fit_cas_apo <- lm(percentage ~ heat * time, data = cas_apo)
anova_cas_apo <- car::Anova(fit_cas_apo, type = 2)
emm_cas_apo <- emmeans(fit_cas_apo, ~ heat | time)
tukey_cas_apo <- pairs(emm_cas_apo, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "Caspase", state = "Apo")
letters_cas_apo <- cld(emm_cas_apo, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "Caspase", state = "Apo")

cas_inoc <- confocal_data %>%
  filter(marker == "Caspase", state == "Inoc") %>%
  droplevels()

fit_cas_inoc <- lm(percentage ~ heat * time, data = cas_inoc)
anova_cas_inoc <- car::Anova(fit_cas_inoc, type = 2)
emm_cas_inoc <- emmeans(fit_cas_inoc, ~ heat | time)
tukey_cas_inoc <- pairs(emm_cas_inoc, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "Caspase", state = "Inoc")
letters_cas_inoc <- cld(emm_cas_inoc, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "Caspase", state = "Inoc")

cas_sym <- confocal_data %>%
  filter(marker == "Caspase", state == "Sym") %>%
  droplevels()

fit_cas_sym <- lm(percentage ~ heat * time, data = cas_sym)
anova_cas_sym <- car::Anova(fit_cas_sym, type = 2)
emm_cas_sym <- emmeans(fit_cas_sym, ~ heat | time)
tukey_cas_sym <- pairs(emm_cas_sym, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "Caspase", state = "Sym")
letters_cas_sym <- cld(emm_cas_sym, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "Caspase", state = "Sym")

anova_table <- bind_rows(
  as.data.frame(anova_edu_apo) %>% rownames_to_column("term") %>% mutate(marker = "EdU", state = "

```

```

as.data.frame(anova_edu_inoc) %>% rownames_to_column("term") %>% mutate(marker = "EdU", state = "I")
as.data.frame(anova_edu_sym) %>% rownames_to_column("term") %>% mutate(marker = "EdU", state = "I")
as.data.frame(anova_cas_apo) %>% rownames_to_column("term") %>% mutate(marker = "Caspase", state = "I")
as.data.frame(anova_cas_inoc) %>% rownames_to_column("term") %>% mutate(marker = "Caspase", state = "I")
as.data.frame(anova_cas_sym) %>% rownames_to_column("term") %>% mutate(marker = "Caspase", state = "I")
) %>%
  select(marker, state, term, everything())

# This is the supplemental table that matches the graph
supplemental_tukey_table <- bind_rows(
  tukey_edu_apo,
  tukey_edu_inoc,
  tukey_edu_sym,
  tukey_cas_apo,
  tukey_cas_inoc,
  tukey_cas_sym
) %>%
  select(marker, state, time, contrast, estimate, SE, df, t.ratio, p.value) %>%
  mutate(
    estimate = round(estimate, 2),
    SE = round(SE, 2),
    t.ratio = round(t.ratio, 2),
    p.value = signif(p.value, 3)
  ) %>%
  arrange(marker, state, time)

write_csv(anova_global_table, "~/Documents/GitHub/heating_lacerates_final/tables/TableS10-confocal_anova.csv")
write_csv(supplemental_tukey_table, "~/Documents/GitHub/heating_lacerates_final/tables/TableS11-confocal_anova.csv")

```